THE FRACTIONAL KNAPSACK PROBLEM

- We are given ‘n’ items and a knapsack.
- The item ‘i’ has a weight w_i and a profit p_i the knapsack has a capacity ‘M’.
- We are also allowed to take an item in fractional part.
- If a fraction x_i, $0 \leq x_i \leq 1$ of item i is placed into the knapsack then a profit of pi xi is earned.
- The objective is to fill the knapsack that maximizes the total profit earned.
- The ith item contributes the weight w_i .x_i to the total weight in the knapsack and profit p_i .x_i to the total profit.
- The problem is stated as:

$$\max \sum_{0 \leq i < n} p_i x_i$$

$$s.t.$$

$$\sum_{0 \leq i < n} w_i x_i \leq M$$

$$0 \leq x_i <= 1$$

$$p_i \geq 0, w_i \geq 0, 0 \leq i < n$$

If we have n objects:

1. i is the object
2. x_i is the fraction of the particular object
3. w_i is the weight of the fraction selected
4. p_i is the price of the fraction selected

then:

- $\sum_{1 \le i \le n} p_i x_i$ needs to be maximized
- $\sum_{1 \le i \le n} w_i x_i \le W$
- $0 \le x_i \le 1$
- $1 \le i \le n$

A feasible solution will satisfy equations 2,3,4 but an optimal solution will satisfy equation 1

Algorithm

- If the items are already been sorted into non-increasing order of p[i] /w[i] then the algorithm given below obtains solutions corresponding to this strategy.

```
Algorithm GreedyKnapsack (m, n)
// P[1 : n] and w[1 : n] contain the profits and weights respectively
of  items ordered so that p[i] / w[i] > p[i + 1] / w[i + 1].
// m is the knapsack size and x[1: n] is the solution vector.
{
 for i := 1 to n do x[i] := 0.0 // initialize x
 U := m;
  for i := 1 to n do
   {
```

```
    if (w[i] > U) then break;
    x [i] := 1.0; U := U – w[i];
    }
  if (i < n) then x[i] := U / w[i];
}
```

Algorithm: Greedy-Fractional-Knapsack (w[1..n], p[1..n], W)

```
for i = 1 to n
  do x[i] = 0
weight = 0
for i = 1 to n
  if weight + w[i] ≤ W then
     x[i] = 1
     weight = weight + w[i]
  else
     x[i] = (W - weight) / w[i]
     weight = W
     break
return x
```

Example:

1. Consider the following instance of the knapsack problem: n = 3, m = 20, **(P1, P2, P3) = (25, 24, 15) and (W1, W2, W3) = (18, 15, 10).**

Item-1

Profit=25-p1

Weight=18-w1

P1/w1=25/18=**1.38**

Item-2

P2=24

W2=15

P2/w2=24/15=**1.66**

Item=3

P3=15
W3=10
P3/w3=15/10=**1.5**

2. Find out the optimal solution for the following 3 items (P1, P2, P3) = (500,900,700) and (W1, W2, W3) = (1, 3, 7). Knapsack capacity is 5.

item	Profit(pi)	Weight(wi)	Pi/wi
2	24	15	1.6
3	15	10	1.5
1	25	18	1.38

Solution={0,0,0}

Item2 will be selected (maximum Pi/Wi)

Weight of item 2 is 15

Capacity=20

Solution={0,1,0}

Remaining capacity=20-15=5

Item 3 is next item to be selected

Weight =10

Item 3(fraction) = 5/10=1/2=.5

Solution={0,1,1/2)

Remaining capacity=0

Profit=sum of PiXi=0*25+1*24+.5*15=0+24+7.5=**31.5**

Example

Let us consider that the capacity of the knapsack ***W* = 60** and the list of provided items are shown in the following table −

Item	A	B	C	D
Profit	280	100	120	120
Weight	40	10	20	24
Ratio (Pi / Wi)	7	10	6	5

After sorting, the items are as shown in the following table.

Item	B	A	C	D
Profit	100	280	120	120
Weight	10	40	20	24
Ratio (pi/wi)	10	7	6	5

Solution set={0,0,0,0}

Item B is selected first

Weight=10

Profit=100

10<60=B is selected

Solution = {0,1,0,0}

Remaining Capacity=60-10=50

Item A is selected

Weight=40

Profit=280

40<50=A is selected

Solution={1,1,0,0}

Remaining capacity =50-40=10

Item C is selected

Weight=20

Profit=120

20>10=item c (fraction) can be selected=10/20=1/2

Solution={1,1,1/2,0}

Remaining capacity=10-10=0

$\sum P_i X_{i=} P1x1+p2+x2+p3x3=1*280+1*100+1/2*120+0*120=280+100+60=440$